

Week 1: Decision Theory*

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1 Introduction

In these notes, we briefly cover the basic notions of decision theory. The setting follows: there is a joint distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X}$ is the input or feature space and $\mathcal{Y}$ is the label space.

We aim to find a good predictor, i.e., a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that maps feature vectors x to the corresponding label $y \in \mathcal{Y}$. To mathematically measure the performance of any such predictor, we need to specify a performance metric. We then assume to have a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$. In practice, $\ell(a, b)$ measures the loss when the predictor guesses label a , while the actual label is b .

Example 1.1. *We report here the most common loss functions:*

- The 0-1 loss, common for classification: $\ell(a, b) = \mathbb{1}\{a \neq b\}$
- The squared loss for regression: $\ell(a, b) = (a - b)^2$
- The absolute loss for regression: $\ell(a, b) = |a - b|$

Example 1.2 (Binary Classification). *Consider a binary classification task, i.e., when $\mathcal{Y} = \{0, 1\}$. Any loss for this problem is fully specified by four numbers: $\ell(1, 1)$ (the loss value for true positive TP), $\ell(0, 0)$ (for true negative TN), $\ell(1, 0)$ (for false positive), and $\ell(0, 1)$ (for false negative).*

Note

While the distribution $\mathcal{D}$ is given by the environment, the loss function is a modeling choice of the model designer. Depending on the application at hand, you may want to choose specific loss functions.

Definition 1.3 (Expected Risk). *Given distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{X}$ and loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$, the expected risk of a predictor $f : \mathcal{X} \rightarrow \mathcal{Y}$ is defined as $R(f) = \mathbb{E}[\ell(f(X), Y)]$.*

We remark that the expectation in the definition of risk is with respect to the random sampling of (X, Y) according to $\mathcal{D}$.

*These notes are meant as a teaching support for the students of the “Machine Learning” course for the Master’s Degree in Engineering in Computer Science and Artificial Intelligence that I am teaching in Fall 2025. They are likely full of typos and imprecisions, so please get in touch with me in class if something feels off. From time to time I may update new versions of these notes, so make sure to download the latest version.

Example 1.4. Consider a binary classification task on $\mathcal{X} = \{1, 2, 3\}$, with the 0-1 loss, and a predictor f that maps $\{1, 3\}$ in 1 and $\{2\}$ in 0. We want to compute its expected risk when (X, Y) follows the following distribution:

$$\begin{cases} \mathbb{P}(X = 1, Y = 0) = 0, & \mathbb{P}(X = 2, Y = 0) = 1/2, & \mathbb{P}(X = 3, Y = 0) = 1/12 \\ \mathbb{P}(X = 1, Y = 1) = 1/6, & \mathbb{P}(X = 2, Y = 1) = 1/12, & \mathbb{P}(X = 3, Y = 1) = 1/6 \end{cases}$$

We have the following calculation: $R(f) = \frac{1}{2} \cdot 1 + \frac{1}{12} \cdot 1 = \frac{1}{6}$.

It is then natural to conclude that the best predictor—given a fixed distribution $\mathcal{D}$ and loss function ℓ —is one that minimizes the expected risk.

Definition 1.5 ((Bayes) Optimal Predictor). Given distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$ and loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$, a predictor $f : \mathcal{X} \rightarrow \mathcal{Y}$ is said to be optimal if minimizes the expected risk $R(f)$.

Exploiting the rules of conditional expectation, it is fairly easy to derive a closed formula for the optimal predictor: it maps features to the label that minimizes the loss, *given* the realized feature.

Theorem 1.6. The optimal predictor f^* is defined as follows.

$$f^*(x) \in \arg \min_{z \in \mathcal{Y}} \mathbb{E}[\ell(z, Y) | X = x] \quad \forall x \in \mathcal{X}.$$

Note: the expectation in the definition is *only* with respect to Y , and it is conditioned with respect to $X = x$. Furthermore, the optimal predictor may not be unique, as the arg min could contain more than one element for some x .

Proof. We start by writing down explicitly the expected risk for a generic predictor f :

$$R(f) = \mathbb{E}[\ell(f(X), Y)] = \mathbb{E}[\underbrace{\mathbb{E}[\ell(f(X), Y) | X]}].$$

The predictor f^* minimizes the term underlined *for each* $x \in \mathcal{X}$, therefore it is optimal. □

The expression above may seem mysterious, but it is fairly simple if applied to specific distributions. For instance, if Y is discrete, then it can be rewritten as:

$$f^*(x) = \arg \min_{z \in \mathcal{Y}} \sum_{y \in \mathcal{Y}} \ell(z, y) \mathbb{P}(Y = y | X = x).$$

Similarly, if Y conditioned on $X = x$ admits a probability density function $p_{Y|X=x}$ for any $x \in \mathcal{X}$, then the optimal predictor is

$$f^*(x) = \arg \min_{z \in \mathcal{Y}} \int_{\mathcal{Y}} \ell(z, y) p_{Y|X=x}(y) dy.$$

Example 1.7. The predictor studied in Exercise 1.4 is optimal for that problem.

Important

- Computing the Bayes optimal predictor requires *exact* knowledge of $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$.
- There is an inherent risk that is *unavoidable*! There is generally no *perfect* predictor.

Exercise 1.8. Consider a regression problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$, and the quadratic loss defined as $\ell(a, b) = (a - b)^2$. Compute the optimal predictor.

Exercise 1.9. Consider a regression problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$, and the absolute loss defined as $\ell(a, b) = |a - b|$. Compute the optimal predictor. Hint: Assume that Y conditioned on $X = x$ always admits a density function.

Exercise 1.10. Consider a classification task on k classes with the 0-1 loss. Compute the expected risk of the randomized predictor that outputs a class uniformly at random, independently of the x observed.

Exercise 1.11. Consider the binary classification problem with $\mathcal{Y} = \{-1, 1\}$ and the 0-1 loss. Relate the risk of a predictor f to that of its opposite $-f$.

Exercise 1.12 (Multi-Variate Independent Gaussians). Consider a multi-class classification problem where $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{1, 2, \dots, k\}$, and the loss is the 0-1 loss. We make an assumption on the random variable $X = (X_1, \dots, X_d)$: conditioning on any $Y = j$, X_i is an independent gaussian distribution with mean μ_i^j and standard deviation σ_i^j . Compute the optimal predictor.

2 Binary Classification

In binary classification, the label set $\mathcal{Y}$ comprises only two outcomes: true/false, accept/reject, positive/negative. We use $\mathcal{Y} = \{0, 1\}$ for simplicity. Any loss function for a binary classification problem is thus characterized by four numbers, corresponding to the elements of $\{0, 1\}^2$.

Proposition 2.1. In binary classification with loss ℓ , the optimal predictor is given by

$$f^*(x) = \mathbb{1} \left\{ \mathbb{P}(Y = 1|X = x) \geq \frac{\ell(1, 0) - \ell(0, 0)}{\ell(0, 1) - \ell(1, 1)} \mathbb{P}(Y = 0|X = x) \right\}$$

Proof. Consider any fixed $x \in \mathcal{X}$, we want to understand what the best outcome to predict is. We can compute explicitly the expected loss for the two options we have:

$$\begin{cases} \ell(1, 1) \mathbb{P}(Y = 1|X = x) + \ell(1, 0) \mathbb{P}(Y = 0|X = x) & \text{if we predict 1} \\ \ell(0, 1) \mathbb{P}(Y = 1|X = x) + \ell(0, 0) \mathbb{P}(Y = 0|X = x) & \text{if we predict 0} \end{cases}$$

Therefore, it is optimal to predict 1 if the first equation is smaller than or equal to the second one, and 0 otherwise. The statement of the Proposition follows by rearranging terms. $\square$

Since there are only two possible outcomes for Y , it is easier to write the optimal predictor as follows, applying the definition of conditional probability:

$$f^*(x) = \mathbb{1} \left\{ \frac{\mathbb{P}(X = x|Y = 1)}{\mathbb{P}(X = x|Y = 0)} \geq \frac{[\ell(1, 0) - \ell(0, 0)]\mathbb{P}(Y = 0)}{[\ell(0, 1) - \ell(1, 1)]\mathbb{P}(Y = 1)} \right\}$$

Note, the right-hand-side term appearing in this rewriting of the optimal predictor is *independent* from x , while the left-hand-side term is important enough to get its own name.

Definition 2.2 (Likelihood Ratio and Test). The likelihood ratio is the ratio of the likelihood functions:

$$\mathcal{L}(x) = \frac{\mathbb{P}(X = x|Y = 1)}{\mathbb{P}(X = x|Y = 0)}.$$

A likelihood ratio test is a predictor of the form $\mathbb{1}\{\mathcal{L}(x) \geq \eta\}$, for some scalar $\eta > 0$.

It is often useful not to focus directly on the likelihood function but to apply some monotone function to simplify the calculations. In fact, $\mathbb{1}\{\mathcal{L}(x) \geq \eta\} = \mathbb{1}\{h(\mathcal{L}(x)) \geq h(\eta)\}$, for any monotonically increasing function.

Exercise 2.3. Compute the optimal predictor for the binary classification problem, where the loss is $\ell(a, b) = \mathbb{1}\{a \neq b\}$, and the two labels are equally likely, i.e., $\mathbb{P}(Y = 1) = \mathbb{P}(Y = 0)$.

Exercise 2.4. Compute the optimal predictor for the following binary classification problem. The labels are such that $\mathbb{P}(Y = 1) = 10^{-6}$, while X behaves according to $\mathcal{N}(0, 1)$ if $Y = 0$, or according to $\mathcal{N}(s, 1)$ otherwise, for some $s \in \mathbb{R}$. The loss function is

$$\begin{cases} \ell(0, 0) &= 0 \\ \ell(1, 1) &= -10^6 \\ \ell(1, 0) &= 100 \\ \ell(0, 1) &= 0 \end{cases}$$

Exercise 2.5. Compute the optimal predictor for the following binary classification problem with 0-1 loss. The two labels have the same probability, i.e., $\mathbb{P}(Y = 1) = \mathbb{P}(Y = 0) = 1/2$; X conditioned on $Y = 0$ is a standard 2-dimensional gaussian, while conditioning on $Y = 1$ is a gaussian centered in $(1, 1)$ with covariance matrix $1/2 \cdot \mathbb{I}$ (where $\mathbb{I}$ is the two dimensional identity matrix).

There are only four possible outcomes for a binary classifier f :

	$Y = 0$	$Y = 1$
$f(X) = 0$	true negative	false negative
$f(X) = 1$	false positive	true positive

If we normalize with respect to the corresponding populations, we get the following rates that characterize a predictor:

- **True Positive Rate:** $\text{TPR} = \mathbb{P}(f(X) = 1|Y = 1)$, also called *power*, *sensitivity*, or *recall*.
- **False Negative Rate:** $\text{FNR} = 1 - \text{TPR} = \mathbb{P}(f(X) = 0|Y = 1)$, also known as *type II error* or *probability of missed detection*.
- **False Positive Rate:** $\text{FPR} = \mathbb{P}(f(X) = 1|Y = 0)$, also known as *size*, *type I error*, or *probability of false alarm*.
- **True Negative Rate:** $\text{TNR} = 1 - \text{FPR} = \mathbb{P}(f(X) = 0|Y = 0)$, also known as *specificity*.

Starting from these rates, there are a few metrics that are massively implemented in machine learning:

- **Precision:** $\mathbb{P}(Y = 1|f(Y) = 1)$.
- **F1-score:** F_1 is the harmonic mean of prediction and recall.

$$F_1 = \frac{2\text{TPR}}{1 + \text{TPR} + \frac{\mathbb{P}(Y=0)}{\mathbb{P}(Y=1)}\text{FPR}}.$$

- **False discovery rate:** expected ratio of false positives over the total number of positives.

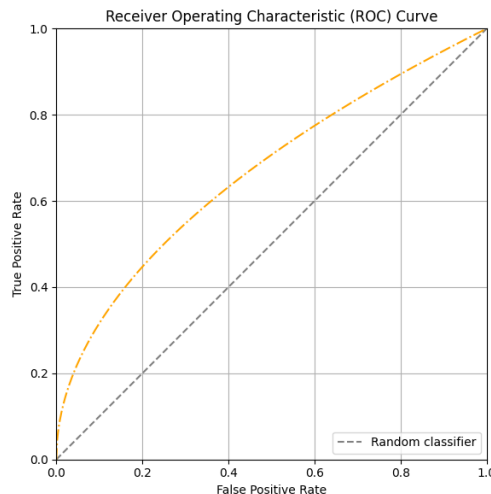


Figure 1: Example of a ROC curve

Suppose we want to maximize the true positive rate subject to a constraint on the false positive rate. Namely, we want to find the predictor that solves:

$$\begin{aligned} \max \quad & \text{TPR} \\ \text{subject to} \quad & \text{FPR} \leq \alpha. \end{aligned}$$

The following Theorem (that we do not prove) states that the right thing to do is to look at the likelihood ratio test.

Theorem 2.6 (Neyman-Pearson Lemma). *Suppose that the conditional densities $f(x|y)$ are continuous. Then the optimal predictor that maximizes TPR with an upper bound on FPR is a likelihood ratio test.*

Finally, imagine that you do not have a clear idea of what the *right* loss function for your problem is, but want to get the *best* out of the problem at hand. In other words, you may want to understand the better trade-off between recall (TPR) and size (FPR). A helpful tool is to *plot* for any $\alpha \in [0, 1]$ what is the best TPR achievable with the constraint that $\text{FPR} \leq \alpha$ in the FPR-TPR plane.

Remark 2.7. *By Theorem 2.6, the ROC curve is given by varying the threshold in the likelihood ratio test from negative to positive infinity.*

Exercise 2.8. *Plot the ROC curve for the classification problem described in Exercise 2.4.*

3 Naive Bayes Classifier

Everything worked so far by assuming *full knowledge* of the joint distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$. This is an unrealistic assumption: *what's the use of learning if we know everything?*

We want to estimate the conditional probabilities (i.e., the ones appearing in the left-hand side of the likelihood ratio $\mathcal{L}(\mathbf{x})$), for feature $\mathbf{x}$, that is a d -dimensional object (either continuous or discrete). Instead of trying to learn $\mathbb{P}(X = \mathbf{x}|Y = y)$ for all possible pairs $(\mathbf{x}, y)$, the Naive Bayes Classifier makes the optimistic (and naive) assumption that all the coordinates $X_1, X_2, \dots, X_d$ of X are independent, conditionally on $Y = y$.

Example 3.1 (Discrete Random Variables). Consider the case in which $Y = \{0, 1\}$, and $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_d$ is a discrete set with $|\mathcal{X}_i| = n$. A priori, we would need to learn $2 \cdot n^d$ numbers, namely all the $\mathbb{P}(X = (x_1, \dots, x_n) | Y = y)$. The naive Bayes classifier assumes that all the coordinates of $\mathbf{x}$ are independent given $Y = y$, therefore:

$$\mathbb{P}(X = (x_1, \dots, x_n) | Y = y) = \prod_{i=1}^d \mathbb{P}(X_i = x_i | Y = y).$$

We must only learn $2 \cdot d \cdot n$ numbers. This is an exponentially smaller number!

Example 3.2 (Continuous Random Variables). Consider a multi-class classification setting where $\mathcal{X} = \mathbb{R}^d$, and the loss function is the 0-1. The problem is that we cannot learn **in full generality** a **continuous** number of points in a naive way! There is a classical statistical way around it: instead of estimating the probability density function directly, we **assume** that the underlying distribution has a specific shape, characterized by a few parameters, and we then estimate such parameters! For instance, we can assume that X given $Y = y$ behaves as a multivariate Gaussian distribution (this is the Gaussian Naive Bayes classifier used on the iris dataset). This means that it is characterized by the covariance matrix $\Sigma \in \mathbb{R}^{2d}$ and mean $\mu \in \mathbb{R}^d$, and thus we would need to estimate $O(d^2)$ numbers. The naive Bayes classifier assumes that the dimensions are independent, so we would only need to learn d variances and d means, for a total of $O(d)$ numbers. Remember: you have already computed the optimal predictor for this case in Exercise 1.12.

Important

The Naive Bayes Classifier is not *one single classifier*, but a **method** to estimate the optimal predictor. In particular, it makes the *naive* assumption that all the X_i are independent, conditionally on $Y = y$. This assumption may bring poor results when the data exhibits strong (and asymmetric) correlations, but it avoids the **curse of dimensionality**.

Pointers

These notes are mainly based on Chapters 2.2 and 2.3 of Bach [2024]. Part of the material is from Chapter 2 of Hardt and Recht [2022] (where it is also possible to find formal proof of Theorem 2.6). See also Chapter 24.2 of Shalev-Shwartz and Ben-David [2014].

References

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